

Observations On The Fractional Boundary Operator Induced By A Fractional Stokes' Theorem

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We begin with a simple conjectural extension of Stokes' Theorem: could it be that the fractional derivative implies a fractal exterior or a fractional boundary?

$$\int_M dw = \int_{\partial M} w ; \quad \int_M d^\alpha w = \int_{\partial^\alpha M} w$$

If the theory of fractional derivatives is so well established, can there exist fractional boundary operators? Since the boundary of the boundary is null, we have;

$$\partial^2 = 0$$

Hence, ∂ , the operator which transforms a region to its boundary, is nilpotent.

As we will show, weak formulae for the volume, length, etc. of boundary regions can be obtained not necessarily from a fractional wedge product, but instead from using standard fractional derivatives in the very definition of the exterior derivative;

$$d^\alpha = \sum_{x_n=x,y,z,\dots} \frac{\partial^\alpha}{\partial x_n^\alpha} \wedge dx_n$$

1 Balls In $\mathcal{R}^3$

Let's *loosely* explore this proposition of 'fractional boundary regions' for the Ball of radius R , with the typical $\left(\frac{d}{dx}\right)^k x^n = nPr(n, k) x^{(n-k)}$;

$$\begin{aligned} \int_{B(0,R)} \nabla^\alpha \cdot [x, y, z] &= \int_{\partial^\alpha B(0,R)} [x, y, z] \\ \left(\frac{1}{\alpha}\right) \alpha! \int_{B(0,R)} x^{1-\alpha} + y^{1-\alpha} + z^{1-\alpha} dV &= \int_{\partial^\alpha B(0,R)} [x, y, z] \cdot n \\ 3 \left(\frac{1}{\alpha}\right) \alpha! \int_{B(0,R)} z^{1-\alpha} dV &= R \int_{\partial^\alpha B(0,R)} \frac{[x, y, z]}{R} \cdot n \\ 3 \left(\frac{1}{\alpha}\right) \alpha! \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \int_{-R}^R (r \cos(\phi))^{(1-\alpha)} r^2 \sin(\phi) dr d\phi d\theta &= R \int_{\partial^\alpha B(0,R)} n \cdot n \\ \frac{3}{R} \left(\frac{1}{\alpha}\right) \alpha! (2\pi) \left(\int_{-R}^R r^{(3-\alpha)} dr \right) \left(\int_0^{\frac{\pi}{2}} (\cos(\phi))^{(1-\alpha)} \sin(\phi) d\phi \right) &= \int_{\partial^\alpha B(0,R)} 1 \\ \frac{6\pi R^{(3-\alpha)} \left(\frac{1}{\alpha}\right) \alpha!}{(4-\alpha)(2-\alpha)} (1 - (-1)^{4-\alpha}) &= |\partial^\alpha B(0, R)| \end{aligned}$$

Although unarguably questionable, this result passes several of the necessary criteria; being nilpotent and correctly establishing the area of the boundary sphere.

$$|\partial^1 B(0, R)| = \frac{6\pi R^2 \cdot 1! \cdot 1!}{3 \cdot 1} \cdot 2 = 4\pi R^2$$

$$\lim_{\epsilon \rightarrow 0} |\partial^{k-\epsilon} B(0, R)| = 0 ; \quad k = -4, -2, 2, 3, 4, \dots$$

2 Hollow Cylinder In $\mathcal{R}^3$

Adopting the more traditional notation of differential forms, consider $\omega = (z - \frac{h}{2}) y dx - (z - \frac{h}{2}) x dy$. Let's examine the cylinder standing upright in z with height h and with a disc of radius R in xy ;

$$\iint_S d^\alpha \omega = \oint_{\partial^\alpha} \omega$$

$$\iint_S d^\alpha \omega = \frac{h}{2} \oint_{\partial_{z=h}^\alpha} R \sin(\theta) dx - R \cos(\theta) dy + \frac{-h}{2} \oint_{\partial_{z=0}^\alpha} R \sin(\theta) dx - R \cos(\theta) dy$$

$$\iint_S d^\alpha \omega = \frac{h}{2} \oint_{\partial_{z=h}^\alpha} R \sin(\theta) R \sin(\theta) - R \cos(\theta) (-R \cos(\theta)) + \frac{-h}{2} \oint_{\partial_{z=0}^\alpha} R \sin(\theta) (-R \sin(\theta)) - R \cos(\theta) R \cos(\theta)$$

$$\iint_S d^\alpha \omega = \frac{R^2 h}{2} \left| \oint_{\partial_{z=h}^\alpha} + \oint_{\partial_{z=0}^\alpha} \right|$$

$$\iint_S \nabla^\alpha \times [(z - h/2)y, -(z - h/2)x, 0] = \frac{R^2 h}{2} \left| \oint_{\partial_{z=h}^\alpha} + \oint_{\partial_{z=0}^\alpha} \right|$$

$$\frac{2}{R^2 h} \iint_S \nabla^\alpha \times [(z - h/2)y, -(z - h/2)x, 0] \cdot [x, y, 0] d\Sigma = |\partial^\alpha S|$$

$$\frac{2}{R^2 h} \int_0^h \int_0^{2\pi} \left(\text{nCr}(1, \alpha) \cdot \alpha! \cdot \left(z - \frac{h}{2}\right)^{(1-\alpha)} \frac{(R \cos(\theta))^2}{R} + \text{nCr}(1, \alpha) \cdot \alpha! \cdot \left(z - \frac{h}{2}\right)^{(1-\alpha)} \frac{(R \cos(\theta))^2}{R} \right) \sqrt{(R \cos(\theta))^2 + (R \sin(\theta))^2} d\theta dz$$

$$\frac{4\pi R}{h} \text{nCr}(1, \alpha) \cdot \alpha! \cdot \left(\frac{h}{2}\right)^{(2-\alpha)} \frac{1 - (-1)^{(2-\alpha)}}{2 - \alpha}$$

Which indeed satisfies the usual;

$$|\partial^1 S| = 4\pi R$$

$$\lim_{\epsilon \rightarrow 0} |\partial^{k-\epsilon} S| = 0 ; k = -4, -2, 2, 3, 4 \dots$$

3 Commentary

One glaring flaw with this presented methodology is that we are measuring $|\partial^\alpha S|$ by dotting with the familiar tangent vectors of simply ∂S . For example, in the case of the cylinder we took the typical tangent vectors and their orientation from standard Stoke's theorem nomenclature. It is interesting to consider whether the space within the fractional boundary is indeed spanned by the normal vectors or tangent vectors of the full ∂^1 space. Perhaps a better notation would be;

$$\|\partial^\alpha M\| = \|\{v \cdot \hat{n} ; v \in \partial^\alpha S, \hat{n} \in \partial S^\perp\}\|$$

Though the formulae of Section 1 and 2 correctly achieve nilpotency at integral α , one is faced with the more questionable property that their output at $\alpha = 0$ is 0; a conclusion untruthful to the known volumes of spheres, cylinders, etc.

It should be noted that a trivial 'interpolation' of;

$$\frac{\sin(\pi x)}{x(1-x)\pi} (c_1 x + (1-x)c_0) = \binom{1}{x} (c_1 x + (1-x)c_0)$$

Is merely the binomial coefficient found throughout the various approaches/methods. (With values of c_0, c_1 at $x = 0, x = 1$ and zeros for all other integral x).

Additionally, the distribution of an m-chain composed of a ‘linear combination’ of cells distributes over integration with a set of surfaces Σ in such a way to permit;

$$\Sigma \cdot \omega = \int_{\Sigma} \omega = \vec{\Sigma}^T \mathbf{M} \vec{\omega}$$

Where $\vec{\omega}$ is composed of simpler forms whose linear combination generates ω . Hence, should a basis of forms and surfaces be given, one could construct an ‘algebra of surfaces’ for which $c_1 S_1 + c_2 S_2 + c_3 S_3$ makes very real sense as the linear combination of integrals $c_1 \int_{S_1} + c_2 \int_{S_2} + c_3 \int_{S_3}$ with respect to ω .

4 Nilpotent Matrices: Cubes on $\mathcal{R}^n$

Let us again consider $\mathcal{R}^3$ wherein we can characterise intervals into unions of 3D regions, 2D surfaces, 1D lines, or 0D points.

The reduction in the dimension of the boundary for cubes follows a simple pattern;

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 6 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 2 & 0 \end{pmatrix} \begin{pmatrix} s^3 \rightarrow \text{Cubes} \\ s^2 \rightarrow 6 \text{ Squares} \\ s \rightarrow 4 \text{ Edges} \\ s^0 \rightarrow 2 \text{ Endpoints} \end{pmatrix} = \partial$$

Call this matrix, which is neither technically correct nor nilpotent, P .

The sense of ‘addition’ is instead the *unison* of sets, with P representing the incremental decomposition of the square sets into lower-order boundaries.

In very loose terms, consider the m-chain defined as follows: Numerically, the least-squares stable point of P , under the stipulation of unit magnitude, turns out to be $[0.1798, 0.11088, 0.44427, 0.88883]$. In even less sensible terms, the ‘boundary’ of the set of approximately $\frac{8}{9}$ of a Point $\cup \frac{4}{9}$ of a Line $\cup \frac{1}{9}$ of a Square $\cup \frac{0}{9}$ of a Cube is exceedingly close to itself. However, this approach fails to capture nilpotency, and instead we remark the simpler;

$$\partial \begin{pmatrix} s^3 \\ s^2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 6/s & 0 \end{pmatrix} \begin{pmatrix} s^3 \\ s^2 \end{pmatrix}$$

From simple dimensional reasoning, we would expect a fractional boundary to be of $n - \alpha$ dimension; not within the discrete measure categories of above. In any case, computing;

$$J^\alpha = \sum_{k \geq 0} (J - 1)^k \text{nCr}(\alpha, k) = \begin{pmatrix} 0 & 0 \\ -\frac{6}{s} \sum_{k \geq 0} (-1)^k \text{nCr}(\alpha, k) k & 0 \end{pmatrix}$$

Interestingly enough, a similar series is obtained from interpolating the sequence of $\{s^3, 6s^2, 0, 0, 0, \dots\}$

$$|\partial^n \text{Cube}(s)| = \lim_{N \rightarrow \infty} \sum_{k=0}^N \text{nCr}(n, k) \left((-1)^{(k+1)} 6ks^2 + (-1)^k s^3 \right)$$

For sufficiently large n , these sums collapse down to a simple binomial theorem of value 0. In the region $0 < n < 1$, the behavior is less determinable.